DBMS EXPERIMENT NO. 01

Entity–Relationship (ER) Diagram for an Indian E-Commerce Platform

AIM

To design an Entity–Relationship (ER) diagram for an Indian e-commerce platform involving Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, and Address entities. The ER diagram represents primary keys, composite and multivalued attributes, weak entities, Product specialization, relationships, cardinalities, and participation constraints.

ER DIAGRAM

The ER diagram represents the following entities, attributes, relationships, and constraints:

- Customer and Address
- Product and Category
- Seller
- Order and OrderItem
- Payment and Delivery
- Product specialization into PhysicalProduct and DigitalProduct
- OrderItem as a weak entity
- Composite key consisting of OrderID and ProductID
- Multivalued attribute PhoneNumbers
- 1:1, 1:N, and N:1 cardinalities
- Participation constraints
- Standard ER notation

DESCRIPTION OF ER DIAGRAM

Entities: Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, and Address are represented as the major entities of the e-commerce system.

Primary Keys: Each strong entity contains a unique primary key used to identify its records.

Composite Key: OrderItem uses a composite key consisting of OrderID and ProductID.

Weak Entity: OrderItem is represented as a weak entity because its identification depends on the related Order and Product entities.

Multivalued Attribute: Customer may have multiple phone numbers, represented using the multivalued attribute PhoneNumbers.

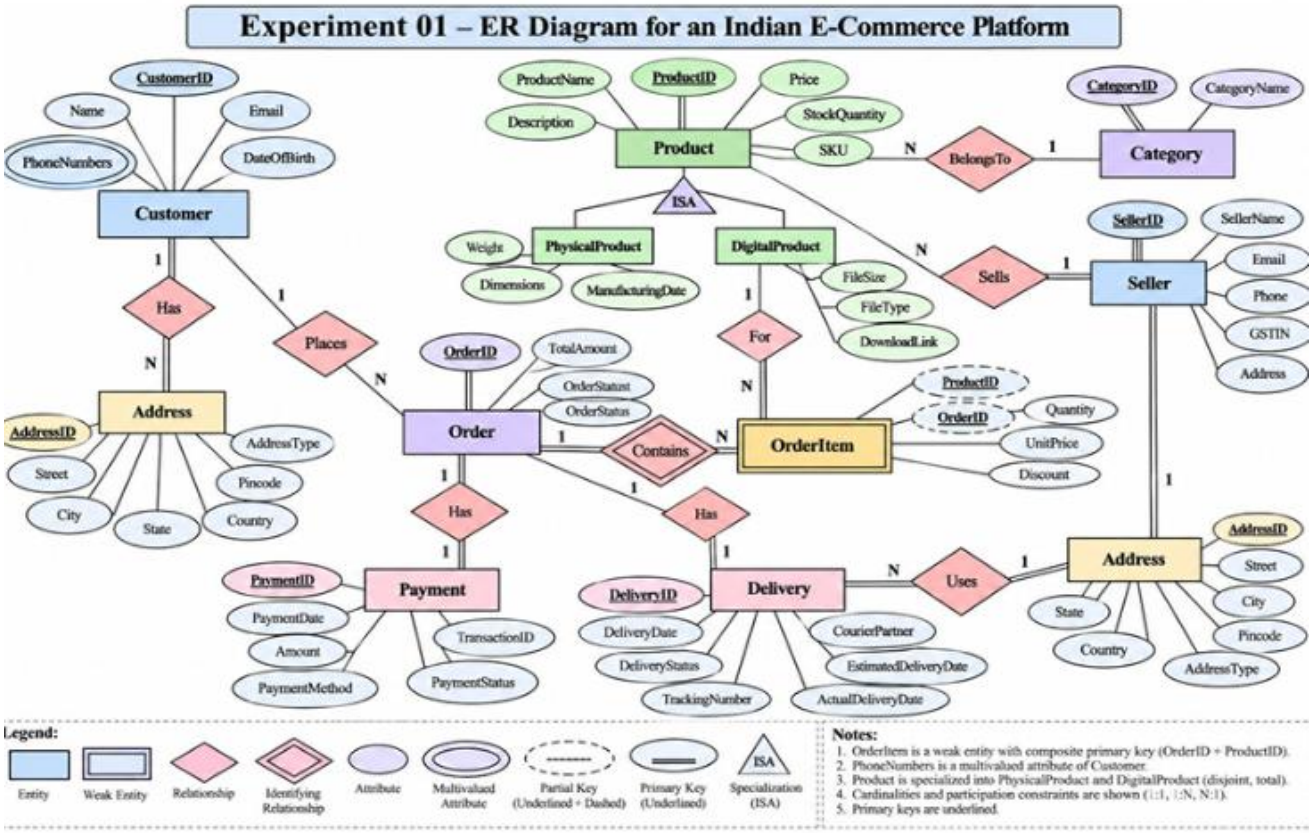
Specialization: The Product entity is specialized into PhysicalProduct and DigitalProduct.

Cardinality: The relationships use 1:1, 1:N, and N:1 cardinalities wherever appropriate.

Participation Constraints: The diagram represents total and partial participation constraints for relevant relationships.

Relationships: Relationships are established among Customer, Address, Product, Category, Seller, Order, OrderItem, Payment, and Delivery according to the e-commerce workflow.

DIAGRAM



RESULT

The Entity–Relationship diagram for the Indian e-commerce platform was successfully designed. The diagram represents the required entities, primary and composite keys, multivalued attributes, weak entity, specialization, relationships, cardinalities, and participation constraints.